Faculty of Computer Science and Mathematics

Lab Exercise 2

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Overview:

This in-class exercise extends Lab Exercise 2. You will apply tables (including colspan and rowspan), forms (text inputs, radio buttons, checkboxes, drop-down lists and labels) and semantic HTML tags (header, nav, section, article and footer) to solve a realistic problem.

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Create tables with headers and merged cells using colspan and rowspan.

2. Build forms with appropriate input types, correctly linked labels and grouped radio buttons.

3. Structure a web page using semantic HTML tags.

4. Identify and correct errors in tables, forms and semantic structure, and validate code using the W3C Validator.

Best of luck!

Problem A: Faculty Event Page with Schedule and Registration

Your faculty is organising a one-day event, for example a Programming Workshop, a Career Day or a Hackathon. The organising committee needs a simple web page that shows the event schedule and allows students to register. Build the page using HTML only (no CSS). You may choose any event name, date and activities.

Instructions

Create a file named lab2_ExerA.html that contains the following:

1. Semantic Structure

- A `<header>` containing an `<h1>` with the event name and a short tagline.
- A `<nav>` with internal links (e.g. `href="#schedule"`, `href="#register"`, `href="#about"`) that jump to each section of the page.
- At least three `<section>` elements, each with an `id`: About the Event, Event Schedule and Registration.
- Inside the About section, two `<article>` elements, each introducing one speaker or one activity (with an `<h3>` and a `<p>`).
- A `<footer>` with the organiser name and a copyright line using `©`.

2. Event Schedule Table

- A table with `border="1"` and header cells (`<th>`) for Time, Activity, Venue and Person in Charge.
- At least 6 rows of activities from morning to evening.
- Use `colspan` at least once (e.g. a "Lunch Break" row that spans across Activity, Venue and Person in Charge).
- Use `rowspan` at least once (e.g. two consecutive sessions held in the same venue).

3. Registration Form

Create a form with `action="register.html"` and `method="post"` that contains:

- Text inputs for Full Name, Matric Number and E-mail, each with a `<label>` whose `for` matches the input `id`.
- Radio buttons for Year of Study (Year 1 – Year 4). All radio buttons must share the same `name` so only one can be selected.
- Checkboxes for the sessions the student wants to join (at least 3 choices).
- A drop-down list (`<select>`) for Programme or Faculty with at least 4 options.
- A submit button.

4. Comments

- Use comments (`<!-- ... -->`) to explain each section of your code.

Submission

Save the file as lab2_ExerA.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerA.html and Screenshot of validation) through Github

Problem B: Course Feedback Page (Debugging and Extend)

A student helper created a Course Feedback page for CSF3133. The page shows a feedback summary table and a feedback form, but it contains several errors. Some radio buttons can be selected at the same time, clicking some labels does nothing, and the table looks broken. Your tasks are to find and fix the errors, then extend the page.

Given Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
<header>
    <h1>CSF3133 Course Feedback</h1>
</head>
<nav>
    <a href="#results">Results</a> | <a href="#form">Feedback Form</a>
</nav>
<section id="results">
    <h2>Feedback Summary</h2>
    <table border="1">
        <tr>
            <th>Week</th>
            <th>Topic</th>
            <td>Rating</td>
        </tr>
        <tr>
            <td>1</td>
            <td>Introduction to HTML</td>
            <td>4.5</td>
        <tr>
            <td>2</td>
            <td colspan="3">Tables and Forms</td>
        </tr>
        <tr>
            <td>3</td>
            <td>Semantic HTML<td>4.0</td>
        </tr>
    </table>
</section>
<section id="form">
    <h2>Give Your Feedback</h2>
    <form action="submit_feedback.php" method="post">
        <label for="name">Name:</label>
        <input type="text" id="fullname" name="name"><br>
        <label>Overall rating:</label><br>
        <input type="radio" id="good" name="good" value="good">
        <label for="good">Good</label>
        <input type="radio" id="poor" name="poor" value="poor">
        <label for="poor">Poor</label><br>
        <input type="checkbox" id="lab" name="lab" value="yes">
        <label for="labs">I attended all labs</label><br>
        <input type="submit" value="Submit">
    </from>
</section>
<footer>
    <p>Copyright &copy 2026 CSF3133</p>
</footer>
</body>
</html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab2_ExerB.html.

2. Identify and correct all errors. At the top of the `<body>`, write an HTML comment listing each error and the line number, e.g. `<!-- Error 1 (Line 1): missing <!DOCTYPE html> -->`.

4. Validate your corrected file at `https://validator.w3.org`. Your final page must show "No errors or warnings to show".

Part 2. Extend into a Two-Page Website

5. Table: add rows for Week 4 and Week 5. Both weeks cover the same topic "Mini Project", so use `rowspan="2"` on the Topic cell. Add a final row "Average Rating" that uses `colspan="2".`

6. Form: add an E-mail text input with a matching `<label>`, a third radio option "Average", a group of at least 3 checkboxes "Topics I want more practice on", and a `<select>` for Section/Group (at least 3 options).

7. Semantic HTML: add a new `<section id="comments">` titled "Student Comments" that contains two <article> elements, each with an `<h3>` (student name) and a `<p>` (comment). Add a link to this section in the `<nav>`.

8. Use comments to explain each new section of your code.

Submission

Save the file as lab2_ExerB.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerB.html and Screenshot of validation) through Github

Rubric

Table A: Rubric for Additional Lab Exercise 2 – Problem A

Criteria	Description	Marks
Table Creation	Schedule table has <th> headers, at least 6 rows of meaningful data, and correct use of colspan and rowspan (no extra or missing cells).	3
Form Structure	Form contains text inputs with matching <label for> and id, a grouped set of radio buttons (same name), at least 3 checkboxes, a <select> with at least 4 options, and a submit button.	3
Semantic HTML	Proper use of <header>, <nav> with working internal links, <section> (min. 3), <article> (min. 2) and <footer>.	2
Code Organization and Readability	Code is well-organized, properly indented and commented. File is named lab2a.html and passes the W3C Validator.	2
Total		**10**

Table B: Rubric for Additional Lab Exercise 2 – Problem B

Criteria	Description	Marks
Error Identification	At least 7 errors correctly identified with line numbers and listed in an HTML comment.	3
Error Correction & Validation	All errors corrected. Radio buttons work as one group, all labels are linked, and the page passes the W3C Validator.	2
Table Extension	Week 4 and Week 5 rows added with correct rowspan; Average Rating row uses correct colspan. No extra or missing cells.	2
Form Extension	E-mail input with label, third radio option in the same group, at least 3 checkboxes and a <select> with at least 3 options.	2
Semantic HTML and Code Organization	New <section> with two <article> elements and a working <nav> link. Code is indented and commented.	1
Total		**10**

Problem A (Codes & Output)

```
1
2   <!DOCTYPE html>
3   <html lang="en">
4   <head>
5       <meta charset="UTF-8">
6       <title>Programming Workshop 2026</title>
7   </head>
8   <body>
9
10      <!-- Header section: Event name and tagline -->
11      <header>
12          <h1>Programming Workshop 2026</h1>
13          <p>Learn, Code and Build Your Future!</p>
14      </header>
15
16      <!-- Navigation section: Links to page sections -->
17      <nav>
18          <a href="#about">About the Event</a> |
19          <a href="#schedule">Event Schedule</a> |
20          <a href="#register">Registration</a>
21      </nav>
22
23      <!-- Section 1: About the Event -->
24      <section id="about">
25          <h2>About the Event</h2>
26          <p>
27              Programming Workshop 2026 is a one-day event
28              designed to help students improve their programming
29              skills and learn from experienced speakers.
30          </p>
31
32          <!-- Article 1: Introduction to the first activity -->
33          <article>
34              <h3>Speaker 1: Introduction to Web Development</h3>
35              <p>
36                  Dr. Ahmad will teach students the basics of HTML
37                  and how to create a simple webpage.
38              </p>
39          </article>
40
41          <!-- Article 2: Introduction to the second activity -->
42          <article>
43              <h3>Activity 2: Coding Challenge</h3>
44              <p>
45                  Students will work individually to solve
46                  programming problems and improve their
47                  problem-solving skills.
48              </p>
49          </article>
50      </section>
52      <!-- Section 2: Event Schedule -->
53      <section id="schedule">
54          <h2>Event Schedule</h2>
55
56          <!-- Table displaying time, activity, venue and person in charge -->
57          <table border="1">
58              <tr>
59                  <th>Time</th>
60                  <th>Activity</th>
61                  <th>Venue</th>
62                  <th>Person in Charge</th>
63              </tr>
64
65              <tr>
66                  <td>8:00 AM - 9:00 AM</td>
67                  <td>Registration</td>
68                  <td>Main Hall</td>
69                  <td>Organizing Committee</td>
70              </tr>
```

```
71
72              <tr>
73                  <td>9:00 AM - 10:00 AM</td>
74                  <td>Opening Ceremony</td>
75                  <td>Main Hall</td>
76                  <td>Event Director</td>
77              </tr>
78
79              <tr>
80                  <td>10:00 AM - 11:00 AM</td>
81                  <td>Introduction to HTML</td>
82                  <td>Computer Lab 1</td>
83                  <td>Dr. Ahmad</td>
84              </tr>
85
86              <tr>
87                  <td>11:00 AM - 12:00 PM</td>
88                  <td>Hands-on Coding</td>
89                  <td>Computer Lab 1</td>
90                  <td>Dr. Aina</td>
91              </tr>
92
93              <!-- colspan merges three columns into one cell -->
94              <tr>
95                  <td>12:00 PM - 1:00 PM</td>
96                  <td colspan="3">Lunch Break</td>
97              </tr>
98
99              <tr>
100                 <td>1:00 PM - 3:00 PM</td>
101                 <td>Group Programming Challenge</td>
102                 <td>Computer Lab 2</td>
103                 <td>Mr. Daniel</td>
104             </tr>
105
106             <tr>
107                 <td>3:00 PM - 4:00 PM</td>
108                 <td>Project Presentation</td>
109                 <td>Main Hall</td>
110                 <td>All Participants</td>
111             </tr>
112
113             <tr>
114                 <td>4:00 PM - 5:00 PM</td>
115                 <td>Closing Ceremony and Awards</td>
116                 <td>Main Hall</td>
117                 <td>Event Director</td>
118             </tr>
119         </table>
120     </section>
122     <!-- Section 3: Registration Form -->
123     <section id="register">
124         <h2>Registration Form</h2>
125
126         <!-- Form collects student information and session choices -->
127         <form action="register.html" method="post">
128
129             <!-- Text input for full name -->
130             <label for="fullname">Full Name:</label>
131             <input type="text" id="fullname" name="fullname" required>
132             <br><br>
133
134             <!-- Text input for matric number -->
135             <label for="matric">Matric Number:</label>
136             <input type="text" id="matric" name="matric" required>
137             <br><br>
138
139             <!-- Email input -->
140             <label for="email">E-mail:</label>
141             <input type="email" id="email" name="email" required>
142             <br><br>
143
```

```
            <!-- Radio buttons: Students can select only one year -->
            <p>Year of Study:</p>

            <input type="radio" id="year1" name="year" value="Year 1" required>
            <label for="year1">Year 1</label>
            <br>

            <input type="radio" id="year2" name="year" value="Year 2">
            <label for="year2">Year 2</label>
            <br>

            <input type="radio" id="year3" name="year" value="Year 3">
            <label for="year3">Year 3</label>
            <br>

            <input type="radio" id="year4" name="year" value="Year 4">
            <label for="year4">Year 4</label>
            <br><br>

            <!-- Checkboxes: Students can select multiple sessions -->
            <p>Choose the sessions you want to join:</p>

            <input type="checkbox" id="html" name="sessions" value="HTML">
            <label for="html">Introduction to HTML (Dr. Ahmad)</label>
            <br>

            <input type="checkbox" id="coding" name="sessions" value="Coding">
            <label for="coding">Hands-on Coding (Dr. Aina)</label>
            <br>

            <input type="checkbox" id="challenge" name="sessions" value="Challenge">
            <label for="challenge">Group Programming Challenge (Mr. Daniel)</label>
            <br>

            <input type="checkbox" id="presentation" name="sessions" value="Presentation">
            <label for="presentation">Project Presentation (Necessary for All Participants)</label>
            <br><br>

            <!-- Drop-down list: Students select their programme or faculty -->
            <label for="faculty">Programme or Faculty:</label>
            <select id="faculty" name="faculty" required>
                <option value="">-- Select Programme or Faculty --</option>
                <option value="Computer Science">Computer Science</option>
                <option value="Information Technology">Information Technology</option>
                <option value="Software Engineering">Software Engineering</option>
                <option value="Multimedia">Multimedia</option>
            </select>
            <br><br>

            <!-- Submit button -->
            <button type="submit">Register Now</button>

        </form>
    </section>

    <!-- Footer section: Organization name and copyright -->
    <footer>
        <p>Organized by the Faculty of Computing</p>
        <p>&copy; 2026 Programming Workshop. All rights reserved.</p>
    </footer>

</body>
</html>
```

Programming Workshop 2026

Learn, Code and Build Your Future!

About the Event | Event Schedule | Registration

About the Event

Programming Workshop 2026 is a one-day event designed to help students improve their programming skills and learn from experienced speakers.

Speaker 1: Introduction to Web Development

Dr. Ahmad will teach students the basics of HTML and how to create a simple webpage.

Activity 2: Coding Challenge

Students will work individually to solve programming problems and improve their problem-solving skills.

Event Schedule

Time	Activity	Venue	Person in Charge
8:00 AM - 9:00 AM	Registration	Main Hall	Organizing Committee
9:00 AM - 10:00 AM	Opening Ceremony	Main Hall	Event Director
10:00 AM - 11:00 AM	Introduction to HTML	Computer Lab 1	Dr. Ahmad
11:00 AM - 12:00 PM	Hands-on Coding	Computer Lab 1	Dr. Aina
12:00 PM - 1:00 PM	Lunch Break		
1:00 PM - 3:00 PM	Group Programming Challenge	Computer Lab 2	Mr. Daniel
3:00 PM - 4:00 PM	Project Presentation	Main Hall	All Participants
4:00 PM - 5:00 PM	Closing Ceremony and Awards	Main Hall	Event Director

Registration Form

Full Name:

Matric Number:

E-mail:

Year of Study:

- ○ Year 1
- ○ Year 2
- ○ Year 3
- ○ Year 4

Choose the sessions you want to join:

- ☐ Introduction to HTML (Dr. Ahmad)
- ☐ Hands-on Coding (Dr. Aina)
- ☐ Group Programming Challenge (Mr. Daniel)
- ☐ Project Presentation (Necessary for All Participants)

Programme or Faculty: -- Select Programme or Faculty --

Register Now

Organized by the Faculty of Computing

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Problem B (Codes & Output)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>

    <!-- Header section -->
    <header>
        <h1>CSF3133 Course Feedback</h1>
    </header>

    <!-- Navigation section -->
    <nav>
        <a href="#results">Results</a> |
        <a href="#form">Feedback Form</a> |
        <a href="#comments">Student Comments</a>
    </nav>

    <!-- Feedback summary table -->
    <section id="results">
        <h2>Feedback Summary</h2>

        <table border="1">
            <tr>
                <th>Week</th>
                <th>Topic</th>
                <th>Rating</th>
            </tr>

            <tr>
                <td>1</td>
                <td>Introduction to HTML</td>
                <td>4.5</td>
            </tr>

            <tr>
                <td>2</td>
                <td>Tables and Forms</td>
                <td>4.2</td>
            </tr>

            <tr>
                <td>3</td>
                <td>Semantic HTML</td>
                <td>4.0</td>
            </tr>
            <!-- Week 4 and Week 5 share the same topic -->
            <tr>
                <td>4</td>
                <td rowspan="2">Mini Project</td>
                <td>4.5</td>
            </tr>

            <tr>
                <td>5</td>
                <td>4.5</td>
            </tr>

            <!-- Average rating row -->
            <tr>
                <td colspan="2">Average Rating</td>
                <td>4.34</td>
            </tr>
        </table>
    </section>

```

```html
    <!-- Feedback form section -->
    <section id="form">
        <h2>Give Your Feedback</h2>

        <form action="submit_feedback.php" method="post">

            <!-- Name input -->
            <label for="fullname">Name:</label>
            <input type="text" id="fullname" name="name" required>
            <br><br>

            <!-- Email input -->
            <label for="email">E-mail:</label>
            <input type="email" id="email" name="email" required>
            <br><br>

            <!-- Overall rating radio buttons -->
            <p>Overall rating:</p>

            <input type="radio" id="good"
                   name="rating" value="good">
            <label for="good">Good</label>

            <input type="radio" id="average"
                   name="rating" value="average">
            <label for="average">Average</label>

            <input type="radio" id="poor"
                   name="rating" value="poor">
            <label for="poor">Poor</label>
            <br><br>
            <!-- Lab attendance checkbox -->
            <input type="checkbox" id="lab"
                   name="lab" value="yes">
            <label for="lab">I attended all labs</label>
            <br><br>

            <!-- Topics for additional practice -->
            <p>Topics I want more practice on:</p>

            <input type="checkbox" id="html"
                   name="practice[]" value="HTML">
            <label for="html">HTML</label>
            <br>

            <input type="checkbox" id="tables"
                   name="practice[]" value="Tables and Forms">
            <label for="tables">Tables and Forms</label>
            <br>
            <input type="checkbox" id="semantic"
                   name="practice[]" value="Semantic HTML">
            <label for="semantic">Semantic HTML</label>
            <br><br>

            <!-- Section or group selection -->
            <label for="group">Section/Group:</label>

            <select id="group" name="group">
                <option value="1">Group 1</option>
                <option value="2">Group 2</option>
                <option value="3">Group 3</option>
            </select>
            <br><br>

            <!-- Submit feedback -->
            <input type="submit" value="Submit">

        </form>
    </section>
```

```
    <!-- Student comments section -->
    <section id="comments">
        <h2>Student Comments</h2>

        <!-- First student comment -->
        <article>
            <h3>Student A</h3>
            <p>The HTML lessons are easy to understand
               and useful for learning web development.</p>
        </article>

        <!-- Second student comment -->
        <article>
            <h3>Student B</h3>
            <p>I would like more practice on tables,
               forms and semantic HTML.</p>
        </article>
    </section>

    <!-- Footer section -->
    <footer>
        <p>Copyright &copy; 2026 CSF3133</p>
    </footer>

    <!-- Errors:
    Error 1 (Line 10): </head> should be </header>.
    Error 2 (Line 20): Rating must use <th>, not <td>.
    Error 3 (Line 26): Missing </tr> for Week 1.
    Error 4 (Line 28): Incorrect colspan="3" in the Week 2 row.
    Error 5 (Line 32): Missing closing </td> for Semantic HTML.
    Error 6 (Line 40): Label for="name" does not match id="fullname".
    Error 7 (Lines 42-45): Radio buttons have different names.
    Error 8 (Line 47): Label for="labs" does not match id="lab".
    Error 9 (Line 49): </from> should be </form>.
    Error 10 (Line 48): The original form is not properly closed.
    Error 11 (Line 52): Copyright entity should be &copy;.
    -->

</body>
</html>
```

CSF3133 Course Feedback

Results | Feedback Form | Student Comments

Feedback Summary

Week	Topic	Rating
1	Introduction to HTML	4.5
2	Tables and Forms	4.2
3	Semantic HTML	4.0
4	Mini Project	4.5
5		4.5
Average Rating		4.34

Give Your Feedback

Name:

E-mail:

Overall rating:

Good Average Poor

I attended all labs

Topics I want more practice on:

HTML
Tables and Forms
Semantic HTML

Section/Group: Group 1

Submit

Student Comments

Student A

The HTML lessons are easy to understand and useful for learning web development.

Student B

I would like more practice on tables, forms and semantic HTML.

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